



NAUVAL ZULFIKAR

Applied ML / NLP Engineer · Responsible AI candidate Bandung, Indonesia (relocatable to Darmstadt, TBD) zulfikar.nauval1998@gmail.com · +44 7300 469048 · +62 821 2567 2264 [LinkedIn](#) · [GitHub](#) · [Portfolio](#) **Visa status:** UK Graduate Visa (active). Indonesian passport. Eligible for EU Researcher (Hosting Agreement) visa for European PhDs.

PROFILE

MSc Business Analytics graduate with a First Class degree from Aston University, specialising in NLP and ML applications for supply chain optimisation. My dissertation, supervised by Dr. Ammar Al-Bazi, focused on blockchain-integrated LLM analysis for supply-chain customer reviews, laying the groundwork for advanced research in multi-document reasoning and AI for scientific evidence synthesis.

EDUCATION

Aston University, Birmingham, UK (*MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024*)

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan (*BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019*)

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship
- Distinctions: Business Data Analysis, Consumer Behaviour, Marketing Research

RESEARCH-RELEVANT PROJECTS

LLM-Generated Adaptive Shipper Decision Rules (2026)

- Developed an end-to-end RAG system using gpt-4o-mini, enhancing decision rules from traveller profiles with a behavioural literature knowledge base.
- github.com/nauvalZulfikar/LLM-Shipper-Profiles

CV-JD Suitability Checker (2024)

- Implemented a DeBERTa-v3 based semantic matching system, outperforming lexical ATS baselines.
- Full ML pipeline: training, evaluation, serving.

MSc Dissertation Pipeline (2024)

- Fine-tuned DeBERTa-v3 on multi-tier supplier review data for multi-aspect classification, integrated into a decision system.
- Graded 1:1.

Crowd-Shipping ABM Behavioural Sweep (2026)

- Conducted a 450-run parameter sweep to evaluate reward and supply-density effects on acceptance rates.
- github.com/nauvalZulfikar/Crowd-Shipping-ABM-

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung (*Data Scientist & Project Manager · Sep 2022 – Present*)

- Automated ETL pipelines and dashboards, achieving 45% faster project delivery and 20% data-efficiency gain.

Syncwell, Birmingham, UK (*Marketing Data Scientist · Jul – Sep 2024*)

- Implemented predictive ML and personalisation strategies, increasing CTR by 40% and follower growth by 22%.

PCOS Challenge, Birmingham, UK (*Strategic Data Scientist Specialist · Jun – Aug 2024*)

- Developed customer attribution models, reducing manual workload by 80%.

Bank Muamalat Indonesia, Jakarta (*Lead Data Scientist · Nov 2019 – Aug 2022*)

- Developed NLP models for credit analysis, achieving a 5.3% investment return.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Comparative evaluation of multiple ML regressors for retail sales forecasting accuracy.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3 fine-tuning, gpt-4o-mini API, RAG, sentence-transformers, classification, regression, time-series forecasting.

Languages/Infrastructure: Python, SQL, R, PySpark, NumPy, Scipy, pandas, scikit-learn, FastAPI, Streamlit.

Mathematical/Formal: MILP (Pyomo, scipy.optimize.milp), Agent-Based Modelling (mesa), Discrete-Event Simulation (simpy), Digital Twin architecture, Bayesian methods.

Governance: Model validation, audit-grade documentation, model staleness monitoring, A/B testing, AWS, Docker, Git, ETL/orchestration.

LANGUAGES

Indonesian (*Native*) · **English** (*Fluent, IELTS 7.0 overall*) · **Japanese** (*Intermediate*)

REFEREES

Dr. Ammar Al-Bazi

Senior Lecturer, Aston Business School, Aston University

Email: a.al-bazi@aston.ac.uk

Dr. Viktor Pekar

Lecturer, Aston Business & Social Sciences

Email: v.pekar@aston.ac.uk